

# Table of contents

|                                                              |   |
|--------------------------------------------------------------|---|
| Electro-Magnetic Field (force caused by potential) . . . . . | 1 |
|--------------------------------------------------------------|---|

## Electro-Magnetic Field (force caused by potential)

- static: localized source charge at rest, only spatial dependency, no time dependency
- stationary: source charge at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistic: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movement
- vacuum vs. matter
- local vs. global of  $\mathcal{B}$ , singularity vs. regularity of  $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

## Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := \text{rot}(A) \stackrel{\text{Matter}}{=} \mu_0 (\mathcal{H} + \mathcal{M}) \stackrel{\text{Vacuum}}{=} \frac{1}{c} \left( \frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left( c p_{\text{dipol}} + m_0 \right) \quad (1)$$

## Magnetic Field in Matter

Excitation  $\hat{=}$  Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

## Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservativ}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[ \underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|}\right) \cdot \frac{v}{c}\right]}{r^2}}_{\text{acceleration}} \right] \quad (3)$$

## Electric Field in Matter

Displacement  $\hat{=}$  Exitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Exication:} \quad \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \stackrel{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \langle \mathcal{E}_m \rangle \quad (4)$$

## Abbreviations

$\Phi$  = electric potential       $A$  = magnetic potential       $j = \nabla \times \mathcal{H}$  = elec. current density

$\mathcal{E}$  = Electric Field       $\mathcal{B}$  = Magnetic Field       $\nabla \cdot \mathcal{B} = 0$        $\mathcal{D}$  = Displacement density of dielectricum       $\mathcal{M}$  = magnetization

$\mathcal{P}$  = Polarisation       $\mathcal{H}$  = magn. exitation, magn. field strength ("magn. displacement")

$n$  = spacial molecule density in volume

$\alpha = 4\pi r^3 = \text{constant}$        $\chi$  = elec. susceptibility of dielectricum       $\kappa = \frac{\mu - \mu_0}{\mu_0}$  = magn. susceptibility

$\langle p_m \rangle$  = spacial average of magn. dipol moment       $\langle \mathcal{E}_m \rangle$  = spacial average of electric field

$\bar{\mathcal{E}}_m$  = time average of electric field       $\mu$  = magn. permeability       $\mu_0$  = magn. constant ...

$\varepsilon$  = dielectricity constant       $\varepsilon_0$  = ...